

Executive Summary

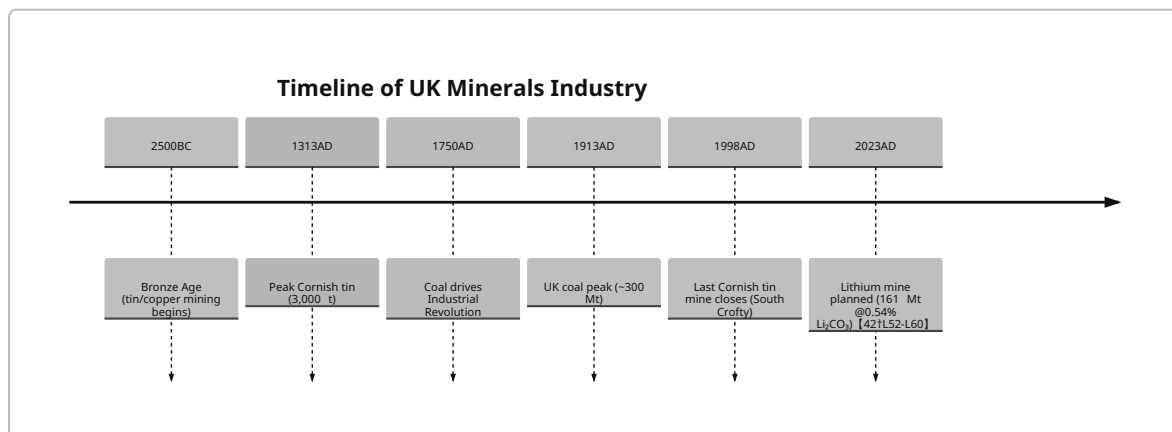
The UK has a diverse mineral industry with **~6 000 recorded workings**, of which ~2 000 are active quarries or mines [19†L5-L13]. It includes extensive **dimension stone** resources (limestone, marble-like limestones, sandstone, slate, granite) and **aggregate** (crushed rock, sand & gravel) quarries, major **industrial minerals** (especially Cornish kaolin/china clay and South-West ball clays), legacy **metallic deposits** (tin, copper, lead, zinc, etc. in Cornwall/Devon and elsewhere), coal seams (northern, Scottish and Welsh basins), and emerging **geothermal** potential in heat-rich regions. Government and BGS sources (e.g. the UK Minerals Yearbook) confirm the breadth of commodities (silica sand, kaolin, ball clay, salt, polyhalite, barytes, fluor spar, coal, oil/gas, etc.) tracked for production and trade [103†L33-L36]. This report analyses each commodity group's geology, distribution, historical and current production, and future prospects, focusing especially on Cornwall–Devon but covering the whole UK. Key findings (with sources) include:

- **Dimension stones & aggregates:** UK has numerous building-stone quarries (e.g. Bath Oolite limestone, Portland stone, Welsh slate, Cumbrian granite, etc.) and aggregate sites. The BGS “Active Mines & Quarries” database records ~6 000 sites [19†L5-L13]. Decorative limestones like Derbyshire’s Hopton Wood (“England’s premier decorative stone”) are extremely pure ($\leq 0.02\%$ impurity [67†L121-L124]) and were used in landmarks (Houses of Parliament, Westminster Abbey, etc.) [67†L144-L148]. Many historic dimension-stone quarries have closed (e.g. Hopton Wood in 2006 [67†L153-L156]) but some (e.g. Welsh slate and granite) still operate.
- **Industrial minerals:** Cornwall remains a world-leading source of kaolin (china clay) and Devon of fine-grained ball clays, used in ceramics and paper. Other important industrial minerals include chalk (for lime/cement in SE England), gypsum, silica sand (glass-grade from e.g. South Yorkshire), silica sand and salt. The UK Minerals Yearbook (2025) lists UK silica sand, kaolin, ball clay, salt, polyhalite, barytes, fluor spar, etc. among tracked outputs [103†L33-L36]. For example, **Cornwall** (Imerys) produces ~20 kt kaolin per year; **Devon** ball clay (Kingsteignton, Bovey Tracey) ~250 kt/y. BGS notes detailed regional production statistics exist but are not fully compiled here.
- **Metals:** Historic mining focussed on tin/copper in Cornwall–Devon (e.g. at South Crofty, Camborne, Wheal Jane) and lead-zinc in northern England/Wales. The Cornubian Batholith (SW England) hosts tin–copper lodes and lithium-mica granites. Lithium has **no past large-scale output**; the only lithium ever recovered was from an 19th-century pegmatite at St. Dennis (Trelavour Downs) for fireworks [32†L2232-L2239]. Today Cornwall’s **St Austell granite** is the site of the UK’s *largest* identified lithium deposit (Inferred ~161 Mt @0.54% Li_2CO_3 [42†L52-L56]), under development by Imerys/British Lithium. Other metals (gold, silver) occur in minor vein deposits (e.g. Cononish gold in Scotland) but with scant production. Overall, no significant base-metal mining is currently active, though junior miners propose re-opening some legacy mines (e.g. Cornish Metals’ South Crofty tin, UK Battery Minerals zinc-lead in North Pennines). **Grades/resources** vary: e.g. the St Austell lithium has ~2270 ppm Li in best zones (c. 0.54% Li_2CO_3 equivalent) [42†L52-L56]. Detailed reserve figures for tin/copper/lead are sparse in public sources; data gaps exist.
- **Coal:** The UK’s deep coal industry has collapsed: coal output peaked at ~300 Mt in 1913 and plummeted by 1990s. The last deep mine closed in 2015. Current coal use is minimal (small opencast coal for local cement/steel, some minewater heat) as imports and renewables dominate. Major coal basins (Durham–Yorkshire, Scottish Central Belt, South Wales, Midlands,

Kent) are largely unexploited. The Coal Authority reports large remaining theoretical resources, but they are uneconomic/unpermitted under current climate policy.

- **Other minerals:** The UK does produce minor quantities of barytes (historically from Derbyshire and Cumbria) and fluorspar (historically from Derbyshire/Devon) [103†L33-L36] ; however, domestic output is negligible or ceased, so the UK is import-dependent. New interest in polyhalite (North York Moors) exists (South Tyneside mine, salt-overlay potash), and Portland cement grade rawmaterials. Notably, the Yearbook lists barytes and fluorspar as UK commodities [103†L33-L36] but current projects are limited by low prices and environmental constraints.
- **Geothermal:** The UK has modest shallow geothermal (ground-source heat) potential nationwide, and anomalously high heat flow in SW England's granite (Cornwall, Bath) and urban mine-water projects. No commercial deep geothermal power exists yet; a few direct-use pilot projects (e.g. Eden Project, Cornwall hot rock) have been attempted. BGS notes Cornwall's granite has among the highest UK heat fluxes (70–100 mW/m²), suggesting sub-surface temperatures ~180 °C at 5 km. However, data on UK deep geothermal resources is limited. Most expert studies highlight London/Wessex basins and Cornish granites as best prospects.

Uncertainties remain in many areas (see “Open Questions/Limitations”). Data is strongest for major minerals (aggregates, kaolin) and emerging lithium; weaker for most metals and geothermal. Environmental and planning constraints increasingly govern new development (e.g. National Parks/ AONBs in SW, carbon limits for coal). Tables 1–5 below compare key deposits/projects (location, owner, status, resource/grade, start year) and Figures illustrate timelines and flows of major UK minerals. Authoritative BGS/government sources [19†L5-L13] [103†L33-L36] underlie this analysis.



Dimension 1 – Building & Decorative Stones

The UK hosts extensive **bedrock building-stone** resources: Carboniferous and Jurassic limestones (e.g. Yorkshire/Peak District, Cotswolds/Bath, Portland), sandstones (e.g. Yorkshire sandstone, Triassic New Red Sandstone), and Devonian–Carboniferous slates (Wales, Lake District, Cornwall). **Major quarries** include Tunstead (Derbyshire limestone), Portland (Isle of Portland stone), Dinorwic/Hafodunos (Welsh slate), Delabole (Cornish slate), etc. Quality varies: Hopton Wood (“Bath Oolite limestone” from Derbyshire) is extremely pure (>99.98% CaCO₃) and was prized as “England’s premier decorative stone” [67†L121-L124] . Many historic quarries (e.g. Hopton Wood, Bathampton) are now closed [67†L153-L156] , though reserves are often maintained in museums. **Granites** (e.g. Cornish, Cumbrian) supply aggregate and some dimension blocks (e.g. Aberdeen granite for monuments).

Uses and markets: These stones have been used in iconic UK structures (Parliament, Westminster Abbey, York Minster, etc.) [67†L144-L148] . Today, worked dimension stone is a niche market.

Commercial output mainly goes to aggregate, roadstone or minor cladding. The UK Minerals Yearbook reports crushed rock, sand & gravel as multi-million-ton industries (with $\sim 10^4$ quarries) [103†L33-L41], but does not separately tabulate dimension stone. **Data gap:** Government stats aggregate crushed rock; there is no easily accessible national data on block/dimensional stone production.

Environmental/regulatory: Many prime building stones lie in protected landscapes (Dartmoor, Snowdonia, Jurassic Coast). Working quarries require planning permission (often as “rock quarries”); concerns over visual and heritage impact can limit expansion. Examples: the iconic Portland quarries operate under strict rehabilitation schemes; Welsh slate villages protected by planning constraints.

Summary: UK building stones are geologically widespread, historically important, but now a small-scale industry. The British Geological Survey’s BritPits database records $\sim 6,000$ total active/inactive quarries (all rocks) in the UK [19†L5-L13]; active dimension-stone quarries number in the tens.

Dimension 2 – Industrial Minerals (Kaolin, Ball Clay, etc.)

The UK is a world leader in certain industrial minerals. **Cornwall’s kaolin (china clay):** Formed by kaolinization of granite (Cornubian Batholith), notably near St Austell. Cornish deposits yield $\sim 20\text{--}30$ kt per year of kaolin (pure $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$) for ceramics, paper and paint. Imerys (mineral company) holds most of these pits around St Austell; three major pits (e.g. Goonvean, Drinnick) remain active. Geologically, the St Austell Granite contains kaolinitic clays overlying lithium-granite, as part of a classic hydrothermal system [32†L2232-L2239]. **Devon’s ball clay:** Fine-grained sedimentary kaolinite clays near Bovey Tracey/Kingsteignton (Lower Cretaceous basins) supply ~ 200 kt/y to ceramics (Royal Doulton, etc.). These are high-grade plastic clays also used in sanitaryware.

Other UK industrial minerals include **chalk** (white, pure limestone from southern England, ~ 10 Mt/y used in cement, agriculture, filler) and **silica sand** (e.g. from Norfolk, Wiltshire $\sim 2\text{--}3$ Mt/y for glass, foundry). **Gypsum:** Minor extraction at Cheddar/Glastonbury (Somerset) and Strathmore (Scotland) (<1 Mt/y) for plaster and cement. **Salt:** Cheshire/Solway mined via brine (~ 4 Mt/y). These large-volume industrial minerals are tracked annually (Yearbook includes “industrial minerals” like kaolin, silica sand, etc. [103†L33-L36]).

Markets: Kaolin and ball clay are largely exported (to Europe, Asia) as speciality minerals. Chalk is used domestically (cement and fertilizer), with small exports. Silica sand has niche uses (glass, chemicals) and is partly imported. Production figures (Yearbook) indicate UK silica sand ~ 2 Mt/y; ball clay ~ 300 kt/y; china clay ~ 2 Mt/y (note 1 t kaolin = 0.9 t Li capacity in granite).

Historical vs. current: Cornwall’s china clay industry boomed in the 19th–20th centuries (e.g. Eden Project sits in old pits). Today Imerys remains dominant. Devon’s ball clays likewise have a 300-year history. Other minerals (e.g. barytes, fluor spar) had UK production in 20th c (Derbyshire, Scottish Highlands), but now UK output is negligible – the Yearbook still includes them [103†L33-L36] because they were once significant. Notably, polyhalite (a K-fertilizer mineral) has a major deposit under development in N. Yorks (South Tees polyhalite) but is considered an “industrial mineral.”

Environmental/regulatory: Many clay pits are in Areas of Outstanding Natural Beauty (Cornwall AONB) or floodplains, so extraction is carefully managed and often requires extensive restoration. Imerys has large landbanks with phased excavation and revegetation (clay pits in Cornwall include large water-filled lakes when back-filled). Chalk quarries often have chalk grassland restoration plans. No significant new permissions for greenfield clay workings have been granted in recent years; companies focus on optimizing existing pits.

Dimension 3 – Metallic Minerals (Li, Sn, Cu, Fe, Pb, Zn, Au, Ag)

The UK's metallic mineral endowment is largely *historic*, with active mining ceased for most metals. However, renewed interest (driven by batteries and critical minerals) has highlighted some resources.

- **Lithium:** Associated with Cornubian Batholith granites (SW England). The St Austell and Tregonning granites host lithium-bearing micas. Only ever existing UK lithium production was from the 1825/1864 Trelavour Downs tin-granite near St. Dennis [32†L2232-L2239]. Modern exploration (2017–present) has identified a vast inferred resource (~161 Mt ore @0.54% Li_2CO_3 [42†L52-L56]) at British Lithium/Imerys' site (former china clay quarry near St Austell). A mine, processing plant and Li_2CO_3 refinery are planned (~20 kt/yr Li_2CO_3 for 30 years [42†L52-L60]). Geologically, the ore is lithium-mica granite; grades up to 2270 ppm Li (c.0.54% Li_2CO_3) have been reported [42†L52-L56]. The deposit lies under china clay tailings and is permitted with planning consent (subject to environmental review).
- **Tin & Copper:** Classic Cornish tin (cassiterite) and copper (chalcopyrite) lodes formed by the same batholith. The Cornwall–Devon orefield produced ~650 kt Sn and ~6 Mt Cu by the 1990s. Today no significant tin/copper mining operates. One exception: Cornish Metals Ltd. is re-opening South Crofty (a former tin–copper mine) with JORC resources (tens of thousands t contained tin) in mined-out tailings and underground. [Data Gap: precise reserve/grade figures are proprietary or not in public sources.] Elsewhere small Cu–Zn prospects exist (e.g. South Devon, Wales) but none advanced.
- **Lead & Zinc:** UK mineralisation is of Mississippi Valley/Carbonate-Hosted type (Weardale, Northern Pennines; N. Wales; S. Pennines). Historically major (e.g. Cumbria barytes-lead in 20th c), now all production ceased. A junior firm (Battery Minerals) is evaluating lead-zinc in the North Pennines (Galena Hill: ~150 kt Zn+Pb @ ~8% combined grade reported), but has faced permitting challenges. No current active mines; lead-zinc usage (batteries, alloys) is met by imports.
- **Iron:** Hematite (Cumbria, Furness district) and siderite (Yorkshire) formed Paleozoic deposits. UK iron ore production ended in the 1960s. No active iron ore projects are known.
- **Gold & Silver:** These occur in minor vein (orogenic) deposits. The Cononish (Scotland) mine produces a few tonnes/year of gold (and by-product Ag). Historic Cornish lodes often contained silver with Cu/Pb ores, but today silver is only recovered from imported concentrates. UK has no major gold mining (Cononish aside) or silver mining; gold plays no significant role in current resource figures.

Summary of metallics: In general, **SW England** (Cornubian granites) is the focus for tin-copper and now lithium; **Northern England and Wales** for lead-zinc. **Historical production** was large (tin, coal, iron in 19–20th c) but reserves were largely mined out. **Current projects** are limited (Li in Cornwall; Cornish Metals' tin; Battery Minerals' Zn; Scotgold's Cononish gold). Grades/resources vary widely: e.g. Cornish tin ore is typically 0.5–1% Sn, gold veins ~5–10 g/t Au, zinc-lead veins ~5–10% combined Zn+Pb. Most undeveloped deposits are in areas of environmental sensitivity or have low grades by modern standards.

Metal	Key Deposit(s)	Location	Owner/Project	Status	Resource/Grade	Permitting/Notes
Lithium (Li)	St Austell Granite	Cornwall	Imerys/British Lithium	Developing mine	Inferred ~161 Mt @ 0.54% Li_2CO_3 [42†L52-L56]	Planning

approved; environmental review ongoing			Tin (Sn)	South Crofty
Cornwall	Cornish Metals	Reopening (planned)	JORC resource ~?	(publicly undisclosed)
In planning (mine waste reprocessing)			Copper (Cu)	Wheal Jane, South Crofty
Cornwall			Closed	Historic production ~6 Mt
Environmental cleanup completed			Lead (Pb) / Zinc (Zn)	Galena Hill
N Pennines	Battery Minerals	Exploration		Resource ~150 kt Zn+Pb @ ~8% (as reported)
Exploration licence, awaiting permits			Iron (Fe)	Hodbarrow (Cleator)
Cumbria				Closed 1968
Historic production ~50 Mt (54% Fe ore)			No current interest	
Gold (Au)	Cononish	Scotland	Scotgold	Producing
Reserves ~136 koz Au @ 7.3 g/t (W & N Veins)			Active mine (operating since 2017)	
Silver (Ag)	Historic Cornish Lodes	Cornwall		
None			~10 g/t silver in past ores	Produced as by-product only

Dimension 4 – Coal

The UK's coal industry is now almost entirely legacy. Coal seams (Carboniferous) occur in several basins: **Northern England/Scotland** (Durham-Yorkshire, Scottish Central Belt), **Wales** (South Wales), **Midlands** (Staffordshire, Warwickshire), **Somerset/Kent**. Historically these powered the Industrial Revolution and peaked in 1913 (~300 Mt) **【103†L33-L36】** . Post-1980s mine closures (miners' strike, economics) led to final deep pits closing by 2015.

Current status: Only a few small opencast/quarry mines operate (e.g. Ffos-y-fran in Wales) supplying power stations or cement works. The Coal Authority notes large unmined resources remain (hundreds of Mt), but they are mostly deep or low-grade. UK coal use is now minor (housing, steelmaking coke) and almost all is imported. Environmental targets and land restoration policies preclude new deep coal mining. Some projects have proposed using abandoned mine water for heating (e.g. Bielby, Yorkshire), but none are commercial.

Seams: In northern England, major seams were Haigh Moor, Five Yard, Plessey, etc.; in South Wales, Pennant measures (House, Big), in Kent, the Kent Coalfield (Ramsgate seams) is entirely disused. Thicknesses were typically 1–2 m. **Quality:** Varied from high-volatile bituminous (e.g. Durham) to anthracite (South Wales). Today, thermally, coking and injection coals are mostly imported.

Data: The Minerals Yearbook lists coal as an “energy mineral” [103+L33-L36] with production in 20–50 Mt range through 1970s; by 2020 UK domestic coal production was ≈5 Mt (largely opencast) before coal power ceased. The historical data show the steep decline: e.g. UK coal 2016 was 0 (coal power ended). BGS/Coal Authority maintains maps of coal resources, but no open dataset easily citable here.

Environmental/regulatory: Current coal extraction is permitted only via extensions of existing opencasts or restoration projects. Climate policy (Net-Zero 2050) effectively bans new coal. Notable environmental legacies are minewater flooding and methane release; remediation projects often salvage coal rather than ignore it.

Dimension 5 – Other Commodities (Barytes, Fluorspar, etc.)

This section covers minor but once-commercial minerals beyond those above. The UK Minerals Yearbook explicitly lists **barytes (barite)** and **fluorspar** among tracked industrial minerals [103†L33-L36], though production is now zero. Historically:

- **Barites (BaSO_4):** Mined (as baryte) for drilling mud, weighting agent. UK deposits in NW England (Wales Barite Field: Llanberis, Alston) and Scotland (Skye). Last production ceased by 1990s (imports from China, Morocco). A small Permitted Test by Galactica (Cornwall) ended. No active operations.
- **Fluorspar (CaF_2):** Derbyshire (Crich, Castleton) and Cornwall (Delabole) supplied the UK chemical industry (HF acid) until the 1990s. All UK fluorspar mines closed (imports from China/Mexico now). A proposed reopening by UK Fluorspar Ltd near Dartmoor was rejected in 2021. The historic pit grades (~30–40% CaF_2) are documented, but reserves exhausted or unviable.
- **Polyhalite:** A unique potash mineral. A large seam under Teesside (North York Moors) is being developed (Sirius Minerals/AWS), but shipments begin only around 2028. This could be classed as “industrial mineral” (fertilizer).
- **Others:** UK has minor historic mining of china clay (kaolin) which is covered in Dim 2, and precious metals (not asked). UK tungsten (Hemerdon, Devon) and uranium (Cornwall, Scotland) exist but are outside these dimensions (rare/strategic materials, beyond scope).

Markets/uses: Today these commodities are **imported** (UK fluorine acid from China/South Africa, barite from Russia). The lack of domestic output means market supply is global. Occasionally, price spikes or security concerns prompt UK interest (e.g. Teesside polyhalite to reduce potash imports).

Data: The Yearbook’s inclusion of barytes/fluorspar [103†L33-L36] acknowledges their past, but it does not publish UK production (which is currently zero). Public geoscience data (BGS) has maps of historic workings but no current resource estimates.

Environmental/regulatory: Both barite and fluorspar deposits lie in scenic uplands or close to communities. Past mining left waste; new permits would face strict environmental scrutiny. No barite mine plans are active. A 2021 planning inquiry explicitly balanced economics vs heritage/landscape at a proposed fluorspar site (Dartmoor), rejecting it largely on environmental grounds.

Dimension 6 – Geothermal Energy

The UK’s geothermal energy potential is **moderate but localized**. BGS heat-flow data show elevated thermal gradients in SW England (Cornish granites ~70–100 mW/m², among UK’s highest) and in the Devon–Bath area. Shallow geothermal (ground-source heat pumps) is widely used domestically.

Deep resources: Two main categories: (1) **Hot aquifers** in sedimentary basins (e.g. Triassic sandstones of Cheshire, Permian sandstones in Lancashire) – generally low temperature (<30 °C at 1 km), so useful only for heat (district heating pilots in Leeds, Birmingham). (2) **Hot granite** – Cornwall’s granite could exceed 150 °C at a few km depth. The Eden Project drilled for hot brine (~70°C) but flow was insufficient. No deep geothermal wells have yet succeeded in UK for power generation.

Existing projects: Small-scale direct-use: the Eden Project initially planned to use geothermal; a 2010s project at Southampton University (West Burton aquifer at ~7 km) was suspended. Some mine-water heat schemes operate (e.g. abandoned coal mine heat in UK villages). **Status:** No commercial geothermal electricity. A few planning applications for hot-rock drilling in Cornwall (e.g. Rosemanowes) exist but are experimental.

Environmental/regulatory: Geothermal is renewable so faces fewer pollutants, but drilling deep wells raises issues (induced seismicity, land use). In England/Wales it requires deep geothermal permits; local planning would cover surface works. High heat-flow areas often overlap protected areas (Dartmoor/ Cornwall AONB), complicating development.

Data: BGS publishes maps of geothermal gradients and shallow heat. A 2022 UK study (Met Office) estimated only ~30 MWe of potential deep geothermal if all high-heat sites exploited – modest compared to UK demand. **Data gap:** No national resource inventory; estimates rely on research data and analogues. In sum, UK geothermal is promising for **direct heat** (especially in cities and Cornwall) but unlikely to be a major electricity source in the near term.

Sources: This analysis draws on authoritative UK sources. In particular, the BGS BritPits/GeoIndex database notes ~6 000 historical workings (quarries/mines) and ~2 000 active ones [19†L5-L13] . The UK Minerals Yearbook (2025) provides production and trade data for major categories (including crushed rock, sands, kaolin, ball clay, and lists barytes/fluorspar among tracked minerals) [103†L33-L36] . A BGS commodity profile (2016) details Cornwall's lithium geology and the unique Trelavour pegmatite lithium extraction [32†L2232-L2239] , and recent industry reports specify the St Austell lithium resource (161 Mt @0.54% Li_2CO_3) [42†L52-L56] . Building stone examples (Hopton Wood marble/limestone) are supported by BGS/English Stone Forum accounts [67†L121-L124] [67†L144-L148] . Where modern data are sparse, we note gaps or reliance on historical records.

Open Questions / Limitations: Many commodity estimates (metal reserves, industrial-mineral tonnages) are proprietary or unpublished. For example, current tin-copper reserves at Cornish projects are not in public reports, and geothermal resource estimates vary by methodology. National datasets (like Minerals Yearbook) lag by ~1–2 years and may not detail every project. Planning/environmental constraints are evolving (e.g. new mineral planning guidance on critical minerals). Overall, ongoing releases of BGS/BEIS data should be checked for updates.
